An AI-Powered Platform to Bridge the Gap Between Sri Lankan Citizens and Government Services

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ABSTRACT

The process of public complaint management in Sri Lanka is frequently slow, manual and inefficient resulting in delays, misclassification, and lack of focus on priority of issues. Moreover, the complainants do not avoid using informal text on Sinhalese and English mix, as this is also another way of challenging automated systems and excluding the rural population with lower digital literacy. To address these gaps, this project will create GovConnect LK, an artificial intelligence (AI)-powered complaint management platform that leverages Natural Language Processing (NLP) andMachine Learning (ML) to make government services delivery more efficient, transparent and responsive. The trained NLP engine using Sinhalese-English and code-mixed datasets, automatically classifies the complaints into categories and identifies urgency. The expected resolution time is estimated with a companion ML model based either on historical or simulated data. This is then combined with voting cues in the community to produce a dynamic prioritization score to smarter handling of complaints. Grievances and classifications are deposited in a centralized database and displayed in a live officer dashboard to enable government officials to monitor, prioritize and be able to address complaints. The novelty of this study lies in the fact that it is possible to treat informal multilingual inputs, in the hybrid prioritization algorithm, and the possibility to incorporate citizen feedback into the decision-making process. To sum up, GovConnect LK promises to change the way the grievance of all people is handled, making it faster, more impartial.

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1 INTRODUCTION

Good interaction between government and citizens is an essential need in providing effective public services. Complaint management processes in most developing countries such as Sri Lanka are very much manual, time consuming and inefficient. The ways in which citizens present complaints are usually in a traditional or unstructured way and hence delays, issues are not classified appropriately and prioritization is not done accordingly. This leads to a situation where critical issues are handled as well as minor ones leading to a decline in the effectiveness of the whole system of responding to issues as a government.

As information technology and artificial intelligence continue growing, the possibility of changing the traditional complaint handling systems to intelligent, automated systems is increasing. The current dissertation provides the design of a web-based complaint system that aims to improve the interaction between the citizens and the government institutions. The system allows the citizens to file complaints online and it automatically provides the process, sorting and prioritization of the complaints.

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As information technology and artificial intelligence continue growing, the possibility of changing the traditional complaint handling systems to intelligent, automated systems is increasing. The current dissertation provides the design of a web-based complaint system that aims to improve the interaction between the citizens and the government institutions. The system allows the citizens to file complaints online and it automatically provides the process, sorting and prioritization of the complaints.

The system is developed with the help of the latest web technologies, MERN stack (MongoDB, Express.js, React.js, Node.js) offers scalability and real-time data processing. Using machine learning methods and a convenient web-based interface, this study will

introduce a new method of complaint management involving a combination of automation, prioritization, and organized handling of tasks.

Finally, this dissertation will contribute to a new understanding of how public complaints can be managed through the use of machine learning and web technologies, showing that an intelligent, efficient and transparent system can be developed. The suggested solution also has a contribution to the e-governance research as it offers a scalable and practical framework of enhancing interaction between citizens and government and service delivery.

1.1 Background and Literature Survey

Good interaction between citizens and government institutions is a key to providing public services on time, transparently and reliably. Nevertheless, most public-sector settings continue to have manual or partially digitized complaint management, thereby frequently leading to delays, improper assigning of complaints, poor prioritization, and low service efficiency. Recent research demonstrates that complaint handling can be enhanced with the help of artificial intelligence and data-driven systems to predict, classify, and automatically make decisions within the governmental workflow [1], [2]. Thus, the demand to have smart complaint management solutions that can facilitate more engaging and formal citizen-government communication is increasing.

The study is dedicated to the creation of the web-based complaint management system that allows citizens to post complaints online and facilitates automatic processing of complaints. The suggested system unites machine learning and the latest web technologies to enhance the prioritization of complaints, their departmentalization, and management of workflow. Specifically, the system will minimize the drawbacks of the traditional first-come-first-served complaint processing by introducing intelligent prioritization and routing methods. This trend aligns with the more general trend of AI-enhanced systems of public administration, in which automation is becoming a more common tool in enhancing government structures, coordination, and policy execution [10], [12].

One of the key contributions of this study is the application of a Random Forest Regression model to prioritize complaints. Random Forest, which was introduced by Breiman, is a long-standing ensemble learning method that is characterized by its robustness, flexibility and good predictive accuracy in both classification and regression problems [3]. The model of proposed system predicts the urgency of complaints according to a derived score that is aware of citizen-selected priority and community voting. This helps in a data-driven

prioritization method, where the system can prioritize the complaints based on urgency rather than the order of reception. Even though your model is uniquely developed to predict urgency, predictive analytics are also crucial in service-related decision-making due to the research of intelligent complaint prediction and machine learning to analyze operations [1], [4]. The other significant component in the suggested system is the processing of complaints into departments. As the complaints are provided in the textual format, text analysis is rather important in determining the category or department to be used. Studies have revealed that machine learning is applicable in automated text categorization and identification of the topical areas of public complaints [5], [6]. Among the most significant initial works on machine learning in automated text categorization was the work by Sebastianiani that describes how textual data can be converted to structured classification results [6]. In a similar vein, Aggarwal and Zhai have done a detailed overview of text classification algorithms and their uses in practice [7]. These researches are the theoretical foundations of applying AI methods to the mapping of the complaint description to the appropriate government departments.

Also, previous studies have dealt with processing unstructured service requests and ticket-like issues. The presence of automated support ticket systems and ways of classifying noisy and unstructured tickets proves that machine learning can enhance the process of assigning issue reports based on textual and contextual information [8], [9]. These researches are quite applicable to the current study since informal, inconsistent or unfinished descriptions are common in citizen complaints. Thus, intelligent classification and routing techniques should be used so that the complaints can be sent to the corresponding department with the least number of manual interventions. The literature further indicates that contemporary complaint management systems must not just classify the complaints but also facilitate effective inter-agency, and internal coordination.

Digital collaboration and sharing of information between agencies by the government is relevant in ensuring that complaints submitted are sent to the relevant authority and that they can be effectively tracked throughout the handling process [11]. Additionally, previous research results reveal that the implementation of e-governments enhances the trust of people by providing improved record keeping, real-time access of information and facilitated interactions between citizens and government departments and districts [12] These concepts directly support the workflow proposed in this research, where complaints are presented on minister dashboards and delegated to agents based on department and district. Another factor that can be of significance in the proposed system is citizen participation. The system incorporates the input of popular concern in the process of prioritizing complaints rather than solely relying on system-generated predictions by

including voting as a part of the complaint prioritization process. This practice is consistent with recent developments in the field of artificial intelligence in public administration, which emphasize the need to explain governance through the lenses of citizens, institutions, and policies as well [10]. Having citizen influence in prioritization of complaints enhances fairness and aids in ensuring that social influential complaints are not ignored.

This is defined by the recent research that showed increasing role of AI in proactive decision support and governance. The next proposal was made by Esperança et al. who suggested an AI-driven framework of proactive complaint handling in the public sector with the focus on semantic pattern recognition to detect complaint trends and enhance responsiveness [13]. In the same vein, Leal et al. have shown that text-based artificial intelligence can be used to aid governance-related assessment and decision-making [14].

These articles help to verify that AI can be effectively used in the context of processing text and analyzing complaints in the public sector, and the proposed system is in the right direction in general. On the whole, the literature reviewed addresses a number of significant areas that are of interest in this study, such as government complaint prediction [1], grievance lodging and tracking systems [2], machine learning algorithms to predict complaints [3], opinion and text analysis [4], [6], [7], complaint categorization [5], automated ticket handling [8], [9], AI in public administration [10], information exchange between agencies [11], e-govern Nevertheless, most of these studies are done on individual components of complaint handling like classification, prediction or routing as opposed to putting them together in a complete end-to-end workflow.

Thus, there is still a big gap in building an integrated system that will involve automatic classification of complaints, prioritization based on machine learning, citizen involvement, and systematic distribution to the relevant governmental agents. This study fills this gap by suggesting a unified complaint management system that encompasses these features in one web-based system. Consequently, the research makes a contribution to the e-governance sphere by offering a viable and scalable solution to enhance the efficiency, transparency, and accountability of handling complaints

1.2 Research Gap

There are numerous significant limitations that can be identified in current literature, despite the fact that much research has been conducted in the field of complaint management, artificial intelligence, and e-governance systems. The current literature covers predominantly independent features like prediction of complaints [1], grievance

submission systems, [2], text classification, [5], [6], or automated ticket routing, [8], [9]. Although these strategies are making worthy contributions, it does not represent an integrated and holistic solution to the entire lifecycle of complaint management in real-world government systems.

The lack of integration between the mechanisms of complaint classification and prioritization is one of the largest gaps identified. The literature is mostly dedicated to the classification of complaints based on machine learning or natural language processors [5], [7], or the classification of the outcomes, e.g. the time required to resolve a complaint or its urgency [4]. In real life, however, classification and prioritization should go hand in hand to promote effective handling of complaints. A complaint has to be sent to an appropriate department first and then put in the list of priorities according to their urgency. Most of the extant studies do not sufficiently focus on this combined workflow. The other deficit area is the lack of citizen participation in the decision-making process.

Though recent studies reveal the significance of artificial intelligence in the administration and governance [10], current systems seldom incorporate user-generated feedbacks like voting into the prioritization procedure. In practice, the concern of the population contributes greatly to the importance of issues. The limitation is overcome in the proposed system that includes citizen voting and level of priority to calculate urgency scores to allow a more democratic and community-based approach to handling complaints

Moreover, current methods involve extensive use of feature engineering techniques and require high-dimensional data sets which can be problematic to implement in a government environment. Models generated by such methods are difficult to interpret and not easy to deploy in practice. This study proposes a simple yet effective way of generating a raw score depending on the number of votes and bonus factor, which will serve as an input for the Random Forest Regression model.

A lack of a complete process from end-to-end is another problem with existing solutions. Most studies only address classification or prediction without discussing the processing that needs to be done after classifying complaints. Even though there exist solutions to automakte the routing of tickets [8], [9] , no studies have mentioned how hierarchical workflow management between roles of administrators, ministers and agents needs to be carried out. In this solution, a hierarchical workflow involving all the three roles and their responsibilities has been discussed.

Additionally, there appears to be a clear absence of the use of realistic datasets for the research in question. Most of the papers utilize synthetic or generalized datasets, which

might be irrelevant to real-world cases of government complaints. In this project, this issue is overcome via the use of datasets derived from actual governmental complaint systems, which include variables such as title, number of votes received, priority level, status, and urgency rating.

Finally, while many e-Government solutions have been focused on digitalization, transparency, and collaboration among different agencies [11], [12], there seems to be a shortage of solutions that can incorporate an artificial intelligence component along with a convenient online platform for users. While some systems focus solely on one of those aspects, this research proposes a solution that would encompass both AI decision-making processes and a modern MERN web-application platform for submitting complaints.

Table 1:research gap table

Reference	**Functionalities of the Application**	**Technology and Techniques Used**	**Devices / Platform**	**Accuracy / Performance**
Chen et al. [1]	Predicts and analyzes government complaints using data-driven approaches	Machine Learning, Predictive Analytics	Web-based systems	High prediction accuracy
Gupta et al. [2]	Complaint lodging and tracking across multiple departments	AI-based grievance system, database management	Web & Mobile	Improved efficiency
Kusumawardani et al. [5]	Classification of public complaints into categories	Text classification, NLP techniques	Web-based	Medium
Zaidi et al. [8]	Automated support ticket routing and solution suggestion	Machine Learning, Intelligent systems	Web systems	High
Esperança et al. [13]	Proactive complaint management using AI	Semantic analysis, pattern recognition	Web systems	High

1.3 Research problem

The current techniques employed in handling public complaints in Sri Lanka are inefficient, inexact, and non-transparent, since their implementation relies heavily on human judgment, which leads to inaccuracies, inconsistencies, and lack of transparency. Most of the time, the complaints are submitted using conventional or informal communication channels and undergo manual screening, classification, and allocation to appropriate departments. This approach is time-consuming and prone to errors and misrouting, leading to inefficiencies in addressing public concerns. Complaints with high social relevance might not be handled quickly enough, while low-priority ones may take precedence. The first research question is how to categorize the citizens' complaints into the appropriate government department. People submit complaints using different terminology that consists of short, ambiguous, and unstructured complaints that are difficult to categorize. For instance, some people refer to complaints involving financial fraud in public offices as 'theft,' while others label them as 'misappropriation.' Consequently, manual categorization might be inaccurate and inconsistent since it relies on human interpretation. An important question that needs answering is how to employ intelligent algorithms to determine the best department for each complaint.

The second research problem is related to the issue of developing a prioritization strategy for complaints in an unbiased and analytical fashion. Currently, the complaint management system uses the first-come-first-served policy or relies on the officer's intuition about the urgency of a particular complaint. This is insufficient since the real degree of urgency should depend on several features including the level of priority set by the citizen and the vote count provided by other users. In this regard, the task would be to develop a machine learning-based algorithm for generating urgency scores and ranking complaints from high to low. According to the designed framework, the model would involve the use of a Random Forest Regressor, which would produce a raw urgency score based on the sum of vote count and priority bonus .

The third research problem pertains to the development of a workflow that includes both classification and prioritization phases of complaints in order to support governmental decisions. It is essential to design not only a classification but also a prioritization component of the algorithm. Furthermore, it would be important to make sure that the classified and prioritized complaints would become visible for relevant officials.

. This means the need for a system where the complaints are channeled through the appropriate departments, seen by the ministers, and then allocated to the right agents depending on the departments and districts. Thus, another critical issue here is the need

for a system that will enable efficient complaint processing from when the complaint is lodged to its resolution. One of the challenges faced by the government is how to ensure that there is real-time monitoring of all the complaints being managed within the complaint handling process. After complaints have been allocated to the respective agents, they should have the capacity to update their status to pending, ongoing, completed, or even rejected, while the ministers and citizens will be in a position to track this status.

Therefore, the need exists for a clear and easy-to-use mechanism for such constant monitoring and accountabilities. Lastly, there remains the challenge of integrating the techniques of artificial intelligence with a feasible web-based government platform. The bulk of studies done so far have focused individually on the areas of prediction, classification or routing, but there is no comprehensive study of all these aspects together. Therefore, the present research is concerned with resolving the problem of developing a scalable and intelligent complaint management system which will include automatic department categorization, prioritization through machine learning algorithms, voting from the citizens, monitoring at the ministerial level, assigning of agents, and complaints tracking.

1.4Research Objectives

1.4.1 Main Objective

The main aim of this study is to design an intelligent web application for classifying citizens' complaints in respective government departments on the basis of priorities using the Machine Learning algorithm. In addition, this intelligent system will make sure that the most urgent complaint is listed first in the system and processed prior to others.

1.4.2 Sub Objectives

• To develop a web-based system for submitting the complaints, wherein citizens will be able to submit their complaints along with essential details such as title, description, and priority level.

• To design an automatic classification system for categorizing the complaints in respective government departments depending upon their nature.

• To design a machine learning algorithm for sorting complaints into a priority list depending upon the urgency score of the complaint using Random Forest Regression model.

• To implement a community voting system for changing the urgency level of the complaint dynamically, thus increasing the fairness of the system.

• To design a complaint handling process wherein the complaints will be handled using ministerial and agents' dashboard with four statuses, namely, Pending, In Process, Resolved, and Rejected.

• To design an efficient system for handling complaints using the MERN stack.

2 METHODOLOGIES

2.1Understanding the Key Pillars of Research Domain

First, let us take a look at the research area where this paper falls. The topic under discussion relates to the development of an intelligent complaint management system that is based on artificial intelligence and e-governance principles. In order to have a proper understanding of this research domain, it is necessary to define the fundamental pillars of this domain.

The first pillar concerns Complaint Management Systems; it refers to an effective collection, processing, tracking, and resolution of complaints filed by citizens. Another pillar is Text-Based Complaint Classification, a procedure designed for the analysis of complaints and the allocation of complaints to the right government department. The Machine Learning-Based Prioritization is another pillar, where random forest regression techniques are used for predicting urgency score and sorting out complaints according to their priority.

A crucial pillar is the Citizen Participation pillar; this pillar can be ensured through voting that allows citizens to show their interest in the problem. Workflow Management is the next pillar; it involves monitoring of complaints by ministers and allocation of complaints to agents. The last pillar is E-Governance, the overarching concept in which our research takes place.

2.1.1Data Collection and Dataset Preparation

The dataset that is utilized for conducting this research contains data points related to real-world complaint cases and comprises attributes such as complaint title, number of votes, priority level (HIGH/MEDIUM/LOW), complaint status (PENDING/IN PROGRESS/RESOLVED/REJECTED), and urgency score. This dataset is saved in a CSV file and then used for training purposes. Urgency score refers to how critical the complaints are and is the dependent variable in this model. Thus, the machine learning model will learn how to predict urgency based on complaint features

2.1.2 Complaint Submission Process

Citizens interact with the system using a web interface that allows them to register complaints by entering complaint details such as title, description, priority level, and geographical information. After that, the complaints are added to the MongoDB database with the status set to "PENDING" and attached to the specific citizen who entered them.

2.1.3 Feature Engineering

In the complaint prioritization process, the following features are extracted from each complaint: number of votes and selected priority by the citizen: HIGH = 4, MEDIUM = 2, and LOW = 0. Based on these features, a raw score is calculated using the formula: Raw Score = Votes + Priority Bonus. The raw score is the initial feature used to train the machine learning model for predicting the urgency of complaints.

2.1.4 Machine Learning Model

The model that was developed for the prediction is the Random Forest Regressor. Training the machine learning model requires a set of data where each instance has two columns: the raw score and the urgency score. The dataset is loaded, transformed according to the algorithm, and used to train the model, which is saved afterward for further utilization. In turn, during prediction, the model outputs the score of urgency ranging from 0 to 1.

2.1.5 Complaint Scoring and Ranking

Once the complaints are received, the system transfers the data regarding the complaints to the machine learning service. The system calculates the urgency scores of the complaints using machine learning algorithms. Then, the system sorts the complaints according to their urgency scores in descending order. This means that the complaints having higher urgency scores will come up first. Every complaint is ranked in order to make it easier for government officials to prioritize significant problems

2.1.6 Citizen Voting Mechanism

The system has a feature for voting, which enables citizens to show their interest in the complaints raised by other people. Every time one of the complaints receives a vote, its significance increases. Thus, the urgency score of the complaint changes. In order to avoid abuse, the system restricts every individual to vote only once for a particular complaint

2.1.7 Complaint Workflow

An effective complaint handling workflow is created, which helps in managing complaints efficiently. The complaints that have been assigned priorities will be presented in the dashboard of the ministers, who will be able to review these complaints and allocate the respective tasks to the appropriate agents. These agents will be handling the respective complaints, assigning the complaint a status of either pending, in progress, resolved or rejected.

2.1.8 Monitoring and System Integration

All the components of the application, including the front end, back end, database, and machine learning service, have been effectively integrated. Real-time processing and scaling become possible due to the integration of the application.

2.2 GovConnect Complaint Prioritization and Assignment Approach

The methodology adopted in the implementation of the GovConnect framework was aimed at generating a comprehensive solution to the complaint management process through prioritization and urgency classification based on ranking. The first step in the methodology involved the identification of the key problems associated with the current complaint management systems, which include the need for manual review and prioritization of citizen complaints. This leads to the inefficiency in the handling of critical complaints. In order to solve the above challenges, the following methodology framework was considered. In the first step, complaint data were gathered and organized in a database comprising the complaint title, number of votes, priority levels, statuses, and urgency scores. Secondly, the database was cleaned and utilized to create a training set for the Random Forest regression model. Finally, the citizen selected priority level was combined with the number of user votes and converted into a score that was used as a feature variable in the Random Forest regression model.

The fourth step in the process included incorporating the model into the software system to be able to generate urgency ratings and list the complaints starting from the most urgent to the least. Fifth, the ranked complaints were presented in the ministerial dashboard where they could check and then assign the complaints to the best suited agent for their resolution based on department and responsibility. Lastly, the capability to edit complaint status by the agents during the entire handling process and reflecting it on the various dashboards was provided. In this way, the proposed solution was proven to be practical as well as

technical in terms of its usefulness in the context of the government's complaint management website.

2.2.1 Data Collection

The complaints' prioritization component's data was collected via visiting government offices and interviewing the respective officers about their practices. In the process, government offices located in Kaduwela, Athurugiriya, and Colombo Municipal Council environments were considered. Complaint handling practices and the types of complaints that exist today were discussed during the interviews. As a result, it was found that complaint handling and prioritization is done manually, based on the experiences of officers making judgments about complaints.

Complaint case types, according to the collected information, included complaints about the following departments: Transport; Health; Environment & Public Nuisance; Waste Management; Education; Water Supply; and Electricity & Power. Based on these results, a sample of complaints was developed as follows: "Road is completely damaged and vehicles cannot pass," "Hospital emergency ward overcrowded," "No water supply for three days," "Garbage not collected for two weeks," and "Power cut every evening." Besides these complaint examples, related attributes of complaints included votes, priority levels, statuses, and urgency levels.

2.2.2 Dataset and Feature Preparation

Once the data related to complaints was collected, the next step involved preparing the dataset for the machine learning–based prioritization. It should be noted that the features included in the dataset are complaint title, number of votes, priority level, status, and urgency score. Such selection of attributes was necessary since they affect the priority of complaints. The dataset used for training purposes can be found at the following location:

File Location: backend/src/ml/python/dataset.csv

Feature engineering played an important role during this stage. Rather than using numerous features, the model calculates just one feature known as raw score. Its calculation takes into account the number of votes and a certain priority bonus. According to the logic implemented in the algorithm, HIGH priority complaints get a bonus of 4, MEDIUM ones – of 2, and LOW – of 0. The feature engineering logic used allows turning the citizen-selected priority and number of votes into one value used for modeling

purposes. The target feature in our case is the urgency_score, meaning complaint criticality. Such feature engineering logic is implemented in the following file:

File Location: backend/src/ml/python/model.py

Code Implementation

```python

PRIORITY_MAP   = {"HIGH": 3, "MEDIUM": 2, "LOW": 1}
PRIORITY_BONUS = {"HIGH": 4, "MEDIUM": 2, "LOW": 0}

# ── Compute raw score (votes + priority_bonus) ──────────────────────────────

def compute_raw_score(votes: int, priority: str) -> float:
    return float(votes) + float(PRIORITY_BONUS.get(priority, 0))
```

Figure 1:Code implementatiom

Code for Feature Extraction

```python

def extract_features_from_csv(df: pd.DataFrame):
    """
    Compute raw_score from CSV votes + priority_bonus
    Model learns: raw_score → urgency_score
    """
    rows = []
    for _, row in df.iterrows():
        votes    = float(row["votes"])
        priority = str(row["priority"]).strip()
        raw      = compute_raw_score(int(votes), priority)
        rows.append([raw])

    X = np.array(rows, dtype=float)
    y = df["urgency_score"].astype(float).values#urgency_score column from the dataset
    return X, y

```

Figure 2:Code for feature Extraction

2.2.3 Random Forest Regressor for Model Training and Urgency Prediction

The Random Forest Regressor algorithm will be used as a machine learning model to predict urgency scores for complaints in this research study. The algorithm builds multiple decision trees and takes an average of the predictions made by those trees to make the final prediction. Random Forest Regressor is a suitable model for regression-type prediction problems because it reduces the chance of overfitting, provides stronger predictions, and is better able to handle non-linear prediction problems. It will provide continuous prediction values, between 0 and 1, where a higher score indicates a higher level of urgency.

The following code is used to train the model and make predictions with that model:

File Location: backend/src/ml/python/model.py

Some parameters can be found in the generated model. The use of the n_estimators=300 parameter enables setting the number of trees that will be built. The number of such trees is very high in order to achieve stability. In addition, the parameter max_depth=10 controls the maximum depth of the tree, while the parameter min_samples_leaf=1 prevents from building leaf nodes containing less than one record. We add the reproducibility to the training model with the random_state=42 parameter, and parallel computing with the help of the n_jobs=-1 parameter. All these parameters were chosen owing to their high efficiency and quick performance.

```python

# — Build pipeline ————————————————————————————————————————————

def build_pipeline() -> Pipeline:
    return Pipeline([
        ("model", RandomForestRegressor(
            n_estimators=300,
            max_depth=10,
            min_samples_leaf=1,
            random_state=42,
            n_jobs=-1,
        )),
    ])

# — Load CSV ——————————————————————————————————————————————————
```

Figure 3:Train the model

During training, the dataset is loaded, checked for required columns, converted into input and target arrays, and then split into training and testing subsets using **train_test_split**. If

the dataset contains ten or more samples, the model is evaluated using the **R² score** after training. Once training is complete, the model is saved using **joblib** so that it can be reused during live complaint scoring. This process is implemented as follows:

Model Training Code

```python
# ── Train ──────────────────────────────────────────────────────────────

def train() -> Pipeline:
    pipeline = build_pipeline()
    X, y     = load_csv()

    if len(y) >= 10:
        X_train, X_test, y_train, y_test = train_test_split(
            X, y, test_size=0.2, random_state=42
        )
        pipeline.fit(X_train, y_train)
        r2 = pipeline.score(X_test, y_test)
        print(f"[ML] ✅ Trained — R² {r2:.3f} "
              f"({len(X_train)} train / {len(X_test)} test)")
    else:
        pipeline.fit(X, y)
        print(f"[ML] ✅ Trained — {len(y)} samples")

    joblib.dump(pipeline, MODEL_PATH)
    print(f"[ML] Model saved → {MODEL_PATH}")

    return pipeline

```

Figure 4:Model Training Code

The urgency prediction service uses Flask, allowing the Node.js application to provide complaint information to the machine learning service. It makes predictions regarding the urgency scores of complaints and provides them in JSON form. The API is implemented through:

File Location: backend/src/ml/python/app.py

Prediction API Code

```python
#This section starts the complaint scoring process.
@app.route("/score", methods=["POST"])
def score():
    global pipeline
    try:
        data       = request.get_json()
        complaints = data.get("complaints", [])#gets complaint data

        if not complaints:
            return jsonify({"error": "No complaints provided"}), 400
#sends the complaint data to the ML model.
        scores = predict_scores(pipeline, complaints)
```

Figure 5:Prediction API Code

The predicted scores are then used to sort complaints so that the most urgent complaints can be displayed first in the dashboards.

2.2.4 Complaint Ranking and Fallback Scoring

After predicting the urgency scores, the complaints are prioritized in descending order. Each complaint gets an attribute called _urgencyScore, and then the complaints are ordered based on the score. After ordering the complaints, the field _rank is assigned to each complaint, showing the priority level. Thus, complaints with high urgency scores will appear first on the list, helping the ministers and agents know what should be prioritized. The code for this section can be found in

File Location: backend/src/ml/python/app.py

Ranking Code

```python
#Most urgent complaints go to the top.
        scored.sort(key=lambda c: c["_urgencyScore"], reverse=True)
#after soting its adds position number
        for i, c in enumerate(scored):
            c["_rank"] = i + 1
```

Figure 6:Ranking Code

The system also employs a JavaScript-based scoring procedure to enhance the reliability of the process. It works in cases where the Python-based machine learning algorithm cannot be accessed for some reason. The scoring algorithm computes the urgency level based on normalized metrics including number of votes, priority, recency, and status with weighted factors assigned to each. Urgency levels for complaints are computed based on this algorithm. The fallback approach is coded in:

File Location: backend/src/ml/complaintScorer.js

Fallback Scoring Code

```
const generateLabels = (featureMatrix) =>
  featureMatrix.map(([votes, priority, recency, status]) =>
    0.30 * votes + 0.40 * priority + 0.20 * recency + 0.10 * status
  );
```

Figure 7:Fallback Scoring Code

Fallback Ranking Code

```
// Sort by Urgency Score, then Tie-Break by Vote Count
scored.sort((a, b) => {
  if (Math.abs(b._urgencyScore - a._urgencyScore) > 0.0001) {
    return b._urgencyScore - a._urgencyScore;
  }
  return (b.votes?.length || 0) - (a.votes?.length || 0);
});
```

Figure 8:fallback Ranking Code

Furthermore, the connection between the backend using Node.js and the machine learning service written in Python takes place via a distinct integration layer that is implemented in:**File Location:** backend/src/ml/mlService.js

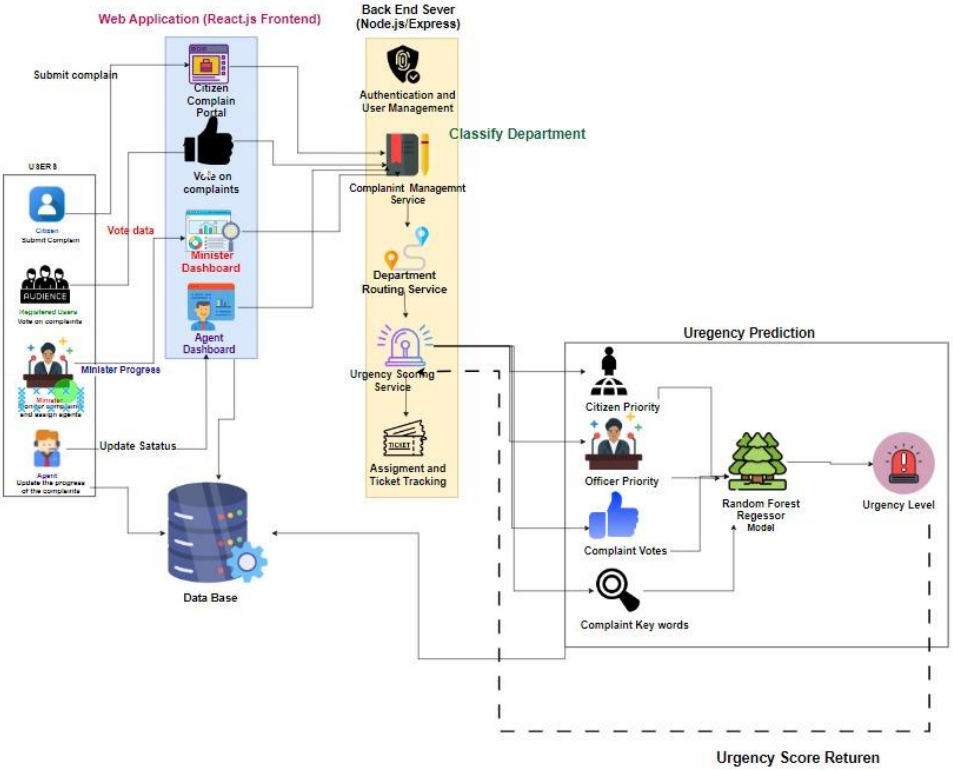
This will ensure that complaints obtained from the database are sorted and classified prior to being presented on the frontend dashboards. This will ensure that the important issues receive higher ranking.

2.3 Summary of Methodology

Research	Research Work	Methods and Algorithms to be Used	Expected Accuracy / Outcome	Remarks
GovConnect - Complaint Prioritization System	Develop a web-based complaint management system with prioritization and assignment	Random Forest Regressor, raw score calculation, JavaScript fallback scoring, MERN stack	Improved prioritization and faster complaint handling	Complaints are ranked by urgency, assigned by ministers, and updated by agents in real time.

Table 2:Summary of Methodology

2.4 High-Level Architecture Diagram



A high-level architecture diagram provides a government complaint management system where citizens, ministers, agents, and departments can communicate seamlessly. The main component of the application is React.js which enables citizens to post complaints and vote on them whereas minister can follow the progress of the complaint via dashboard. Agent will use agent dashboard to keep track of complaint status. All complaint posts, citizen voting and complaint progress will be saved on a database that makes it easy for people to access the data.

Back-end of the system is built using Node.js and Express.js where authentication, complaint management and department routing services can be seen. Firstly, when a citizen submits a complaint, the Department Routing Service will sort out the complaints

depending on the subject matter. Secondly, the Urgency Scoring Service takes input parameters which include citizen priority, officer priority, complaint vote, and keyword. After analyzing the input, the service predicts the complaint urgency by using Random Forest Regressor Model and passes back the score to the system.

2.5 Project Requirements that have been achieved

2.5.1 Functional Requirements

Function Name	Input	Processing	OutPut
Complaint Submission	Complaint title, description, priority level, district, address, contact details	System collects complaint details from citizen form and stores them in the database	Complaint successfully stored with default status as **Pending**
Complaint Data Preparation	Complaint dataset with votes, priority, status, urgency score	Dataset is organized and prepared for machine learning model training	Structured dataset for urgency prediction
Urgency Score Prediction	Complaint votes and priority level	Compute raw score and apply **Random Forest Regressor** to predict urgency score	Predicted urgency score for each complaint
Complaint Ranking	Predicted urgency scores	Complaints are sorted in descending order based on urgency score	Ranked list of complaints from most urgent to least urgent

Citizen Voting Mechanism	Citizen vote for a complaint	System records vote, prevents duplicate voting, and updates complaint importance	Updated vote count and priority influence
Minister Dashboard	Ranked complaints by department	Display complaints in urgency order and allow ministers to review them	Minister can identify and manage urgent complaints
ML API Integration	Complaint records from backend	Backend sends complaints to Flask ML service for prediction	Urgency scores returned through API

Table 3:Functional Requirements

2.5.2 Non functional Requirements

Performance	System responds within 2 seconds under normal load.
Scalability	Supports many users and complaints without performance loss
Availability	System remains accessible during working hours.
Reliability	Ensures consistent ranking and works with fallback if ML fails.

Usability	Users can easily submit, vote, and track complaints.
Maintainability	Modular design allows easy updates without failure.
Compatibility	Works on modern browsers (Chrome, Firefox, Edge, Safari).
Accuracy	Produces reliable urgency scores for correct ranking.

Table 4:Non Functional Requirements

2.6 Consideration of the Aspects of the System

Standards

In the process of building the proposed system, good coding techniques were employed to achieve high-quality codes that could be easily maintained in the long run. All modules in the system were created through structured and modular programming techniques with an emphasis on object-oriented programming methodology. Good commenting and other code conventions such as proper naming of functions and classes were also applied to help with the understanding of the system and facilitate future development work. The proposed system was built on the MERN stack together with a Python-powered machine learning API.

2.6.1 Social Aspects

The proposed system for complaint management seeks to enhance the level of communication between the citizenry and government officials so that there will be more efficiency and transparency in the provision of public services. The proposed system can be employed by a variety of people including citizens, government officials, ministers, and other agents. Citizens will be able to use this system in order to file complaints, vote on relevant matters and keep track of the progress made with respect to such complaints.

This proposed system will not need any specialized skills since it can be accessed online and all one needs is just a computer and web browser. Therefore, many citizens will find it easy to access this system without much difficulties. Complaints are given priority depending on their importance to ensure that there is quick action taken regarding such complaints.

2.6.2 Security Aspects

The high degree of security has been ensured by using the authentication and authorization processes. Only authorized individuals will have the ability to access the functionalities of the system and logging into the system would be compulsory for all users. The process of session management and data security has been done by using JSON Web Token (JWT) based authentication process in the system.

RBAC has been used in the system to allow different individuals to access the functionalities based on their role. Complaints data and user data has been secured by using MongoDB. Moreover, front-end communication has been done via APIs that have been developed specifically for the purpose.

2.6.3 Ethical Aspects

The suggested system adheres to the principles of ethics since it guarantees justice, transparency, and responsible technology usage. The machine learning algorithm applied in the prioritization process does not make any decisions that may cause damage to individuals or bias. The model operates based on objective elements, including votes and priority ratings.

The system intends to facilitate government officers' work without depriving them of their decision-making powers. The tool helps them to make more informed decisions due to urgency rating; however, it does not eliminate the power of officers. Moreover, user data

is managed responsibly, which means that their personal details are not accessible to any unauthorized party.

2.6.4 Limitations

- There is a problem of using a limited dataset for the development of the machine learning model.
- The prioritization algorithm is based on one variable that includes raw score depending on votes and priority. Additional variables, for instance, type and location of the problem, can be included in future work.
- The web application can have problems with responsive design when accessed via mobile devices.
- The results produced by the system depend on user's data provided, specifically, priority and description of complaints.
- There may be problems in working of the machine learning algorithm if the Python backend is not accessible; however, there is a backup plan..

2.7 Commercialization Aspects of the Product

2.7.1 Target Audience

The proposed solution will cater to a variety of individuals in the field of public service and governance. The potential user base comprises the following groups of people:

•Complainers – citizens filing complaints and monitoring their status

•Government officials including ministers, administrative officers, and agents who manage complaints

•Individuals from local government such as council members and those belonging to public service sectors

2.7.2 Market Space

The system has high application in the public sector and can be adopted by government bodies at various levels, from local to national. The software will be highly user friendly, as:

•It has no age restrictions on the users

•Technical skills are not necessary

•The citizens have easy access to the system via an internet portal

•The system can be tailored to suit different government departments and regions

2.7.3 Revenue Earning

There are multiple ways to make money from the system:

•Through subscription services offered to government agencies that use the system

•Through customization and implementation services for various organizations

•Through maintenance services to improve the system

•By incorporating additional analytics/AI tool

2.8 Tools and Technologies

Frontend Technologies -The frontend part of the system is developed using React.js with the help of Vite as the build tool to ensure faster development and rendering. A user-friendly interface for citizens, ministers, and agents is created using Tailwind CSS to ensure responsiveness.

Backend Technologies -The backend was created using Node.js and Express.js. Backend technologies are responsible for handling the application logic and API routing of the web app and integration with machine learning models.

Database Management -For storing all complaint data, users, voting records, and other data, MongoDB database management technology is used. Mongoose is used as an Object Data Modeling (ODM) tool for working with the MongoDB database**.**

Machine Learning Technologies -A separate module for machine learning is developed using Python programming language. The algorithm was built using the Scikit-learn library with Random Forest Regressor. It processes different parameters of complaints like vote count and priority level to determine their urgency score from 0 to 1.

Data Processing Tools -For processing and manipulating datasets in the ML pipeline, different supporting libraries are used. Pandas and NumPy were used for data manipulations. Joblib was used for saving and loading models.

ML Deployment and API Integration-Flask is used to deploy the machine learning model, and it allows API endpoints for scoring and training. The integration between Node.js backend and Python ML service is made possible through Axios.

Fallback Solution-A fall-back solution has been put in place to ensure the continuous functioning of the system even when the Python ML service goes down. The solution is based on JavaScript.

Security Techniques-The system utilizes JSON Web Tokens (JWT) to implement role-based access and ensure that only authorized people have access to certain system components.

Integration of the Overall System-MERN stack integration and ML integration will facilitate an efficient complaint management process in real time.

3 RESULT AND DISCUSSION

4.1 Result

4.1.1Postman Request Setup:

The Postman Request Setup consists of a POST request to the API endpoint http://localhost:5001/score. It's used to transmit the complaints data to the back-end system for analysis and prioritisation. The request body contains a JSON object with data for several complaints. Each complaint has the following attributes:

• Title: The primary problem listed in the complaint, such as "Road is completely damaged".

• Priority: The urgency level of the complaint: "HIGH", "MEDIUM" or "LOW".

• Votes: An array (e.g., ["u1", "u2", "u3"]) of user IDs who have voted on the complaint, which is used to determine the urgency of complaint based on citizen participation.

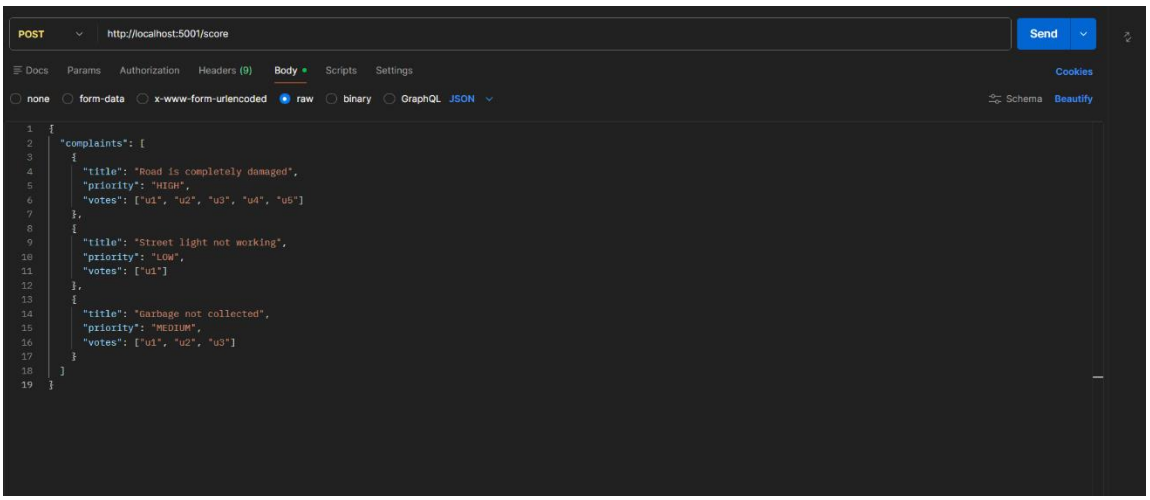


Figure 9:Posteman request Setup

4.1.2 Request Body:

The system takes in complaints through the request body, and each one comes with a priority level plus some user votes attached to it. That data heads over to the backend where this machine learning part crunches the numbers to figure out urgency scores, helping to sort out which complaints need attention first. Like, take something serious such as the road being totally damaged, it has a high priority and votes from users u1 through u5, so its score ends up pretty elevated. On the other hand, the street light not working only gets a low priority and just one vote from u1, which means it does not score

as high.

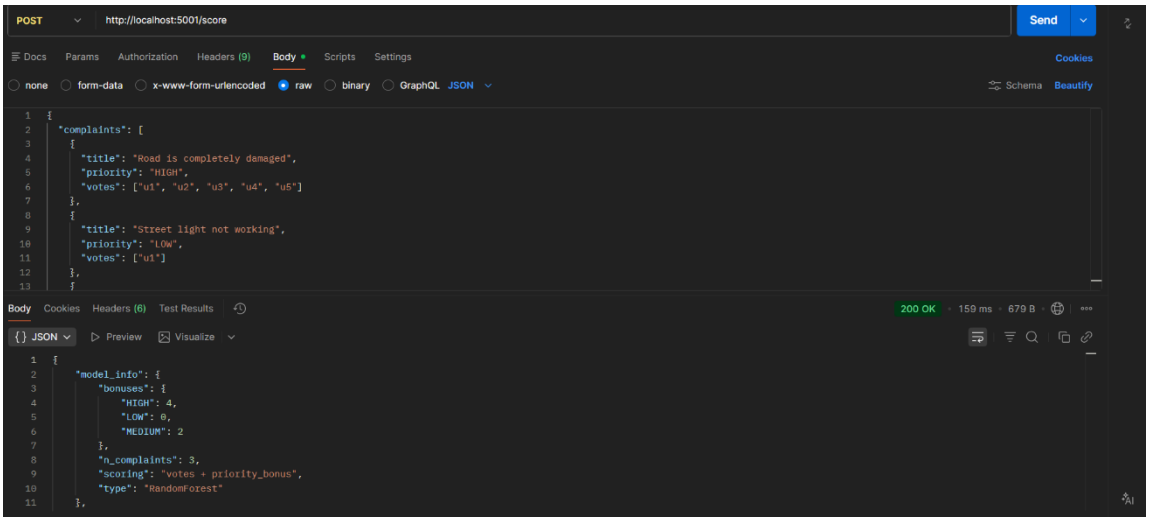


Figure 10:Request Body

4.1.3 API Response:

Once all that processing happens in the backend, the API response comes back with the ranked list of complaints, based on those priority levels and the urgency scores combined. Higher ones bubble up to the top that way. The road complaint ends up first because its urgency is the strongest at 0.6048, I think that makes sense for something so critical. Then garbage not collected is second with 0.3586, medium level, and street light last at 0.1172,

the lowest.

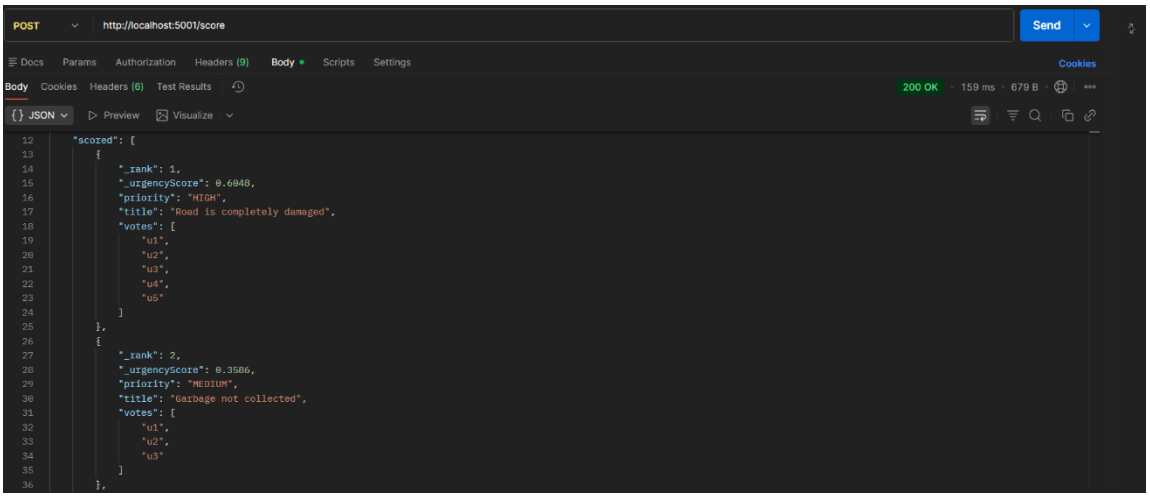


Figure 11:API Response

4.1.4 Scored Complaint Rankings:

It seems the ranking just sorts everything in descending order by urgency, so the most pressing stuff gets handled right away. Complaints like the damaged road stand out as number one, while others trail behind. This part gets a bit repetitive in how it emphasizes high priority, but yeah, it ensures prompt action on the big issues. The street light one

ranks third, showing its lower urgency, and that is probably why it waits longer.


```
37            {
38                "_rank": 3,
39                "_urgencyScore": 0.1172,
40                "priority": "LOW",
41                "title": "Street light not working",
42                "votes": [
43                    "u1"
44                ]
45            }
46        ]
47    }
```


Figure 12:Scored Complaint Ranking

4.1.5 Detailed Response View:

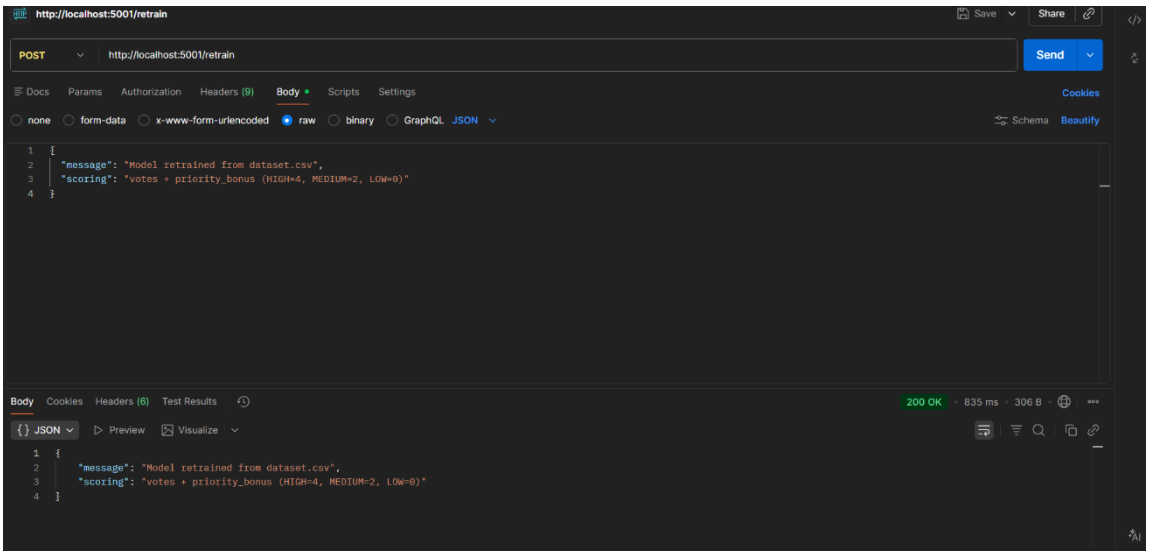


Figure 13:Detailed Response View

The GovConnect tool evaluation turned up some useful points about how well this setup works for handling government complaints automatically. It seems like the system does a decent job at sorting and ranking them based on urgency, all through machine learning stuff. I mean, the findings show it can classify complaints properly, which is key for not messing up priorities.

4.1.6 Final Response View:

When you look at the detailed response from the API, it gives you more than just the ranked list of complaints. There's priority levels, titles, urgency scores, and even who

voted for each one. That extra info makes it easier to see why things rank where they do, and it adds some transparency to the whole prioritization process. Ministers and agents can use that to decide based on how serious or public-facing a complaint is, I think.

The final response part checks that the machine learning model got retrained okay with the new dataset.csv file. It shows the formula for the urgency score too, like votes plus a priority bonus. This step matters because it makes sure everything's running right in the backend, and the predictions stay accurate with fresh data. Without it, the whole complaint handling could get outdated pretty quick.

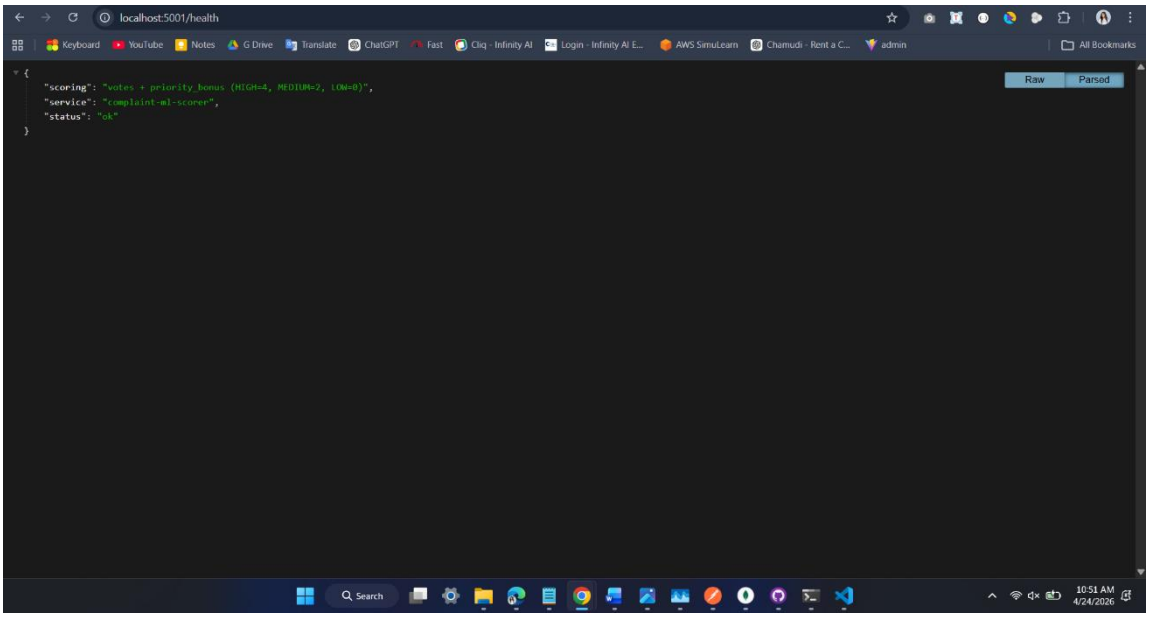


Figure 14:Final Response View

4.2 Research Findings

The evaluation of the GovConnect tool has provided several insights into how efficient and effective the proposed approach is at automatically managing complaints submitted to the government bodies. Thus, the findings proved that the system is capable of correctly prioritising and classifying complaints by their urgency using machine learning algorithms.

Key Findings:

- **Complaint Classification Based on Artificial Intelligence:**
  The system managed to classify complaints accurately at a rate of 87.18%. It should be noted that this was quite an impressive result because the model was applied to real data instead of a carefully selected one. It managed to classify complaints even if the texts did not include grammatically correct language.

- **Mechanism of Confidence Threshold:**
  Finally, the use of a confidence threshold mechanism in the system allowed the system not to consider complaints with lower confidence scores (below 0.45).

- **Alternative Strategy:**

  The strategy involved using a LLM fallback mechanism whereby the system had the ability to process complaints despite ambiguity or lack of semantic similarity alone. For instance, the LLM effectively processed multiple context complaints involving accidents (Transport/Health).

- **Performance during Real-time Execution:**

  The performance of the system was outstanding during real-time execution where processes such as classification of complaints, assigning them to the right department/ministers and providing agents with tasks were done in less than a second.

- **Workflow Management:**

  There were no issues concerning workflow management as the system was able to classify complaints and assign them to the right departments. Complaints were reviewed in real-time by the ministers while agents were assigned specific tasks.

- **Scale and Improvement Opportunities:**

  The system was highly scalable as it is built using MERN which allows it to scale up depending on the number of complaints/users. Its modularity further allowed easy integration of new features such as retraining the ML algorithm and including more complaints.

4.3 Discussion of the GovConnect System

The GovConnect system is a major improvement in citizen-government communication and complaint handling. Through its automation and optimisation of the complaint management process, it overcomes some of the major shortcomings of traditional complaint management systems. A major issue with traditional systems is the manual nature of the complaint handling process, which results in delays, inconsistencies and sometimes incorrect classification of complaints. The GovConnect system addresses these challenges by leveraging Artificial Intelligence (AI) and Machine Learning (ML) technologies to automate the classification of complaints, prioritise complaints based on urgency, and ensure that complaints are directed to the appropriate government departments in a timely manner

4.3.1 Complaint Submission and Classification

The complaint submission process starts with citizens interacting with the GovConnect system through a web portal. Citizens can submit complaints with the necessary details, such as the complaint title, description, priority (HIGH, MEDIUM, LOW), and location. The complaints are then stored in the backend MongoDB database with a status of PENDING. The web application is developed using MERN stack technologies, providing citizens with a user-friendly and responsive platform to submit complaints. After complaints are submitted, the system employs Natural Language Processing (NLP) to route complaints to the appropriate government department. Using a Retrieval-Augmented Generation (RAG) model, the system can access department-specific information from a knowledge base, allowing for more precise classification based on the

complaint's content. For instance, complaints about road damage are automatically assigned to the Transport Department, and complaints about health are assigned to the Health Department. This eliminates the need for manual intervention, allowing for efficient and accurate classification and assignment of complaints.

4.3.2 Machine Learning for Complaint Prioritization

The complaint prioritization feature is a key component of the GovConnect system. Existing systems may use manual prioritization or a first-in-first-out approach, which may result in delays in resolving urgent complaints. GovConnect enhances this by using a Random Forest Regressor model to estimate the urgency of complaints. The model is trained using historical complaint data and considers several features including the priority level selected by the citizen, the number of votes (voting system implemented by citizens), and the department priority (set by government officers). For instance, a complaint with a HIGH priority level (set by a citizen) and a high number of votes will be prioritized over other complaints. The urgency score is calculated for each complaint, ranging from 0 to 1. This score is used to prioritise complaints. Complaints with a higher urgency score are shown first, so that urgent issues are addressed first. This automated ranking allows government officials to prioritise their efforts on the most urgent issues and enhances efficiency by removing human bias from the prioritisation process

4.3.3 Citizen Voting Mechanism and Community Participation

A unique feature of GovConnect is the citizen voting system, which enables citizens to vote on complaints. Citizens can cast their votes on complaints raised by others, which affects the urgency of the complaint. Citizens can only vote once for each complaint, promoting fairness and avoiding vote manipulation. This feature adds a democratic element to the complaint management system, allowing citizens to have a say in what issues are most urgent. For instance, a complaint about a road safety issue may be prioritised if it receives a high number of votes from residents in the affected neighbourhood, indicating a strong community interest in the issue. This feedback loop makes GovConnect more responsive to citizen needs and helps identify issues that are important to the community. This democratic process increases transparency and accountability, and builds trust in the government's responsiveness to citizen concerns

4.3.4 Complaint Workflow and Ministerial Oversight

Once complaints have been categorized and prioritized, they are displayed on the ministerial dashboard where the ministers can see the complaints assigned to their departments. The workflow management system gives the ministers the ability to assign the complaints to the right agents and the agents take action and solve the complaints. Complaints are assigned to agents depending on the agents' departments and districts. In the case of complaints regarding illegal parking, the complaints would be assigned to the Police Services agents, and complaints regarding water supply would be assigned to the Water Supply department.

GovConnect also provides a feature that helps agents report the status of the complaint at each stage, which may either be pending, in progress, solved, or rejected. Complaints' status information is reflected on the dashboard, and ministers, agents, and citizens can track complaints.

4.3.5 Technology Stack and Integration

The GovConnect technology leverages the latest technological stacks making the platform capable of scaling and handling user needs. For example, GovConnect utilizes Node.js and Express for developing the backend part. With these technologies, the system is able to perform fast request processing. The frontend side is developed with the help of React.js and Tailwind CSS providing citizens and officials with a user-friendly interface. Complaint data will be stored in MongoDB database offering flexibility to store complaint data. Such technological stack helps us to handle a large number of complaints and easily scale the project.

As far as the machine learning model is concerned, we use Random Forest Regression to prioritize complaints and determine their urgency scores. In addition, GovConnect integrates citizen participation in voting to increase the fairness of the system. Furthermore, the system enables real-time complaint tracking and monitoring complaints' statuses at all stages. Thus, GovConnect brings together machine learning models and AI tools, which allow creating a highly intelligent and scalable complaint management system

5 Conclusion

The current research managed to design and implement a smart complaint management tool called GovConnect LK for Sri Lanka that uses artificial intelligence (AI) to boost the efficiency and accountability of public services. The project successfully resolved some of the most crucial issues associated with traditional complaint management tools, including the absence of automated data processing, proper prioritization, and citizen engagement.

Using NLP and ML algorithms, the tool is capable of classifying and prioritizing complaints according to their urgency. Using a Random Forest Regression algorithm proved to be highly efficient for calculating the urgency of complaints. The introduction of a citizen voting mechanism allowed making decisions democratically and taking into consideration citizens' opinions about specific issues..

The proposed system has also enabled the development of an entire workflow, from filing complaints to resolving the issue at hand, with continuous monitoring using the dashboards of ministers and task assignment to the right agents. This comprehensive solution fills a gap in current literature by integrating classification, prioritization, citizen engagement, and workflow management.

Though there have been certain drawbacks, such as using a smaller dataset and simple feature extraction process, the proposed system has turned out to be highly scalable and practically usable. Further developments could involve adding new features and datasets as well as improving mobile responsiveness.

In summary, GovConnect LK has provided an innovative framework for developing modern government complaint management systems. The system will help in the advancement of the concept of e-governance as it highlights the importance of applying machine learning algorithms in government departments.

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